Medical and AI Research: A Tale of Two Crises

The Coherence Trap: The Shared Failures of Medical and AI Research

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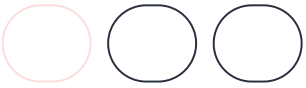

> We live in an age of unprecedented research output, yet the big questions remain: is this true progress, and what is the quality of the work? This article suggests that both medical science and artificial intelligence are trapped in an exponential cycle of growth: a system that rewards the *appearance* of breakthrough over actual discovery. Are we drowning in a tsunami of coherent, plausible, but ultimately hollow papers?
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> Still worse, the rise of Large Language Models (LLMs) is set to escalate an already existing problem. A problem that threatens to collapse the very foundations of knowledge. But maybe there is a way out such as building a more rigorous, truthful, and impactful science for the 21st century.

A Tale of Two Crises

For decades, from working on the front lines of the UK's National Health Service, a certain kind of dissonance has been clear. As a Clinical Scientist and Medical Engineering researcher, one witnessed two parallel realities. In one, the machinery of scientific progress hums along, producing an ever-accelerating stream of research papers, conference presentations, and grant proposals. In the other, on the hospital wards and in the operating theatres, the fundamental metrics of human health, most starkly, life expectancy begins to falter. The upward trajectory is

slowing to a crawl. This is not a future problem; it’s a present-day crisis of knowledge and practice, it’s a real and systemic breakdown infecting two of our most vital scientific disciplines: healthcare and, more recently, Artificial Intelligence.

The explosion of published research, paradoxically coupled with stagnating real-world progress, may not be a sign of a healthy system. It’s a well recognised symptom of a deep-seated rot: a "publish-or-perish" culture that incentivizes the production of superficially "coherent" work, that on closer inspection has substantial failings. Individual studies may appear rational, following the hallowed structure of hypothesis, methods, and conclusion. Yet, collectively, they form a rising tide of information that become an ocean of noise that drowns out the few drops of genuine, impactful knowledge.

And now, Machine Learning (ML), a field once hailed as a potential saviour for medicine's data-rich problems, is not only repeating these same mistakes but is poised to amplify them to a catastrophic degree; specifically with the advent of AI and Large Language Models (LLMs). This article may serve as a diagnosis and a warning, grounded in decades of first-hand experience. Below we chart the alarming parallels between these two fields and examine as crisis that may holding us back and stopping meaningful progress. Maybe the biggest question of all is, can we do something before it is to late?

This analysis will proceed in four parts.

First, it will diagnose the original malady as it has manifested in medical research over the past twenty years. Second, it will demonstrate, with chilling precision, how the nascent field of ML is mirroring this pathology at an accelerated rate. Third, it will examine the role of LLMs as a "crisis multiplier," an engine set to automate the production of scientific chaos. Finally, it will briefly look at other mechanisms we can use to move forward and make real progress.

The Original Sin: Medicine's Lost Decade and the Paper Tsunami

The "Moore's Law of Medical Paper Production"

Around the year 2000, a simple observation began to take shape from within the research departments of the NHS. The sheer volume of medical literature was not just growing; it was expanding at a predictable, exponential rate. Working with a young researcher, Sundip Udani, just beginning his career who had to face this growth, Sundip and myself examined the rate of

growth of medical research. This led to the formulation of a personal rule of thumb, a "Moore's Law of medical paper production": the number of published papers was doubling approximately every 12 years. This was not a casual remark but a quantitative prediction based on the felt reality of trying to keep pace with a field that was, in a very real sense, outrunning itself.

Two decades later, this observation has been borne out by the data with startling accuracy. An analysis of PubMed, the primary repository for biomedical literature, shows that in the year 2000, roughly 485,000 citations were indexed. By 2012, that number had surged to over 811,000, aligning perfectly with the 12-year doubling period. The growth has continued, and while the exponential curve has shown some signs of plateauing, likely due to the finite limits of funding and human researchers, the annual output now consistently exceeds 1.3 million papers.

> The system is producing a city's worth of new research every few months, a volume that no single human, or even a large team, could ever hope to read, let alone critically digest.

The Stagnation Paradox: More Papers, Fewer Gains

This exponential growth in scientific output would be cause for celebration, if it were matched by a corresponding improvement in human health. And that’s a big IF. The stark and sobering reality is that it is not. The very period that saw the most dramatic explosion in medical publications has coincided with a historic stagnation in one of the most fundamental metrics of a society's well-being: life expectancy.

Data from the UK's Office for National Statistics (ONS) and the World Health Organization (WHO) paints a grim picture. After a century of steady gains, improvements in UK life expectancy began to slow dramatically around 2010. Today, it has effectively stalled. The latest

figures for 2021-2023 show a life expectancy at birth of 78.8 years for males and 82.8 years for females. This represents a fall from the pre-pandemic peak and follows a decade of negligible improvement. The Health Foundation, a leading UK charity, has calculated that if the improvement trends seen before 2012 had continued, life expectancy in the UK could have been as much as 3.7 years higher for males and 2.7 years higher for females.

> This disconnect is the central paradox. We are in an age of unprecedented research activity, yet the tangible benefits have diminished. This validates the long-held concern from those within the system that we are, quite literally, "wasting resources and time".

The relationship between the paper deluge and the health stagnation is not a mere correlation; it is a causal feedback loop. The academic system's incentives, particularly the relentless "publish or perish" culture, drive a focus on quantity over quality. This flood of low-impact, often non-reproducible work creates an overwhelming signal-to-noise problem, making it nearly impossible for clinicians and policymakers to identify and implement genuinely effective interventions. Consequently, vast resources funding lab space, and the finite careers of brilliant researchers, are funnelled into the production of more noise, rather than the slow, difficult, and rigorous science that actually makes real progress and saves lives. The system is optimizing for its own internal metrics (papers published, citations accrued) at the direct expense of its societal mission our mission: to improve life and healthcare.

The Reproducibility Crisis: A System Built on Sand

This decline in the quality of this massive research output is no secret. It has a name: the R**eproducibility Crisis**. This is the systemic, well-documented failure of scientists to replicate the findings of their peers, and often, even their own. A landmark 2016 survey published in *Nature*, and corroborated by more recent studies in journals like *PLOS Biology*, revealed the shocking scale of the problem. Over 70% of biomedical researchers reported that they had tried and failed to reproduce another scientist's experiments, and more than half admitted to failing to reproduce their own work.

> Let that sink in for a moment. And then think about all those headlines and results in the media, online, and in the press. All those posts on X, or even on Substack, or on television programs telling us about all the exciting research and how its changing the future - **is it, can it?**

When researchers are asked to identify the cause of this crisis, their answer is overwhelmingly consistent: the "pressure to publish". This then becomes their job and not creating valuable new knowledge as one may think. This pressure forces the rapid publication of studies that are often underpowered, even methodologically flawed, or based on cherry-picked results. It incentivizes novelty over rigor, creating a market for flashy, headline-grabbing claims that are unlikely to stand the test of time.

This directly supports the cynical but accurate assessment: a vast swath of the scientific literature consists not of contributions to human knowledge, but of *career advancement tools*. These are tokens used to secure grants, promotions, and tenure within a broken academic game. The entire edifice of modern medical knowledge is being built on a foundation of sand, with a majority of its bricks liable to crumble under the slightest pressure of independent verification.

> This is, how it is, and we know this. We have measured it

The "Perfect Measurement" Fallacy: A Foundational Flaw

The roots of this crisis run deeper than just perverse incentives. They extend into the fundamental methodological practices of the field. From a medical engineering perspective, one of the most glaring and pervasive flaws is the routine ignorance of **measurement uncertainty**. Unlike in fields like physics, where the rigorous quantification of error is a sacred and non-negotiable part of any experiment, biomedical research often operates under a dangerous fiction: that its measurements are perfect.

This fallacy is everywhere. It is in the analysis of electroencephalograms (EEGs), where the noisy electrical potentials measured on the scalp are treated as a direct window into the brain's software. This is ‘akin to measuring the potentials on the outside of a computer's case to assess what is going on in its software.’

It’s in the interpretation of functional magnetic resonance imaging (fMRI), which measures changes in blood flow as a crude proxy for neural activity in the brain. This is like looking at a PC with an infra-red camera and spotting a hot CPU; and then trying to decode how an LLM is working based on the heat.

These analogies are not hyperbole; they capture the profound disconnect between the tool and the object of study. The results of studies are rightly treated statistically, errors shown, *but not the uncertainty of the measurement itself* - this would be unthinkable in physics.

> This issue is not confined to neuroimaging, that may be a poor example and just one that came to mind when writing. It applies to blood samples, where inter-lab variability can skew results by 20-30%, and to countless other physiological measurements. Yet, these uncertainties are rarely if ever considered in analysis.

This methodological failing is a direct, mechanistic cause of the reproducibility crisis. A scientific study cannot be reliably reproduced if the inherent noise and chaotic variability of its foundational measurements were never properly characterized in the first place.

When a second lab attempts a replication, the researchers will inevitably encounter a different noise profile due to different equipment, a different patient cohort, or even different ambient temperatures. Why? Because the original study failed to model this uncertainty, the replication is almost guaranteed to produce a different result. This isn't merely ‘sloppy science’; its a fundamental failure to grapple with the messy, stochastic reality of biological systems.

> The warning signs from medicine are not just familiar, they are being replayed in AI at a faster tempo.

Déjà Vu: How Machine Learning is Repeating Medicine's Mistakes

The New Moore's Law: An Even Faster Explosion

For those who have spent a career witnessing the slow-motion decline of medical research, the recent trajectory of the Machine Learning field gives rise to a profound and unsettling sense of déjà vu. For me, watching the flood of new papers and the breathless hype cycles, the conclusion is inescapable: I am watching the same issues unfolding in the growth of ML research and papers.

> However, the difference is one of speed. ML is not just repeating medicine's mistakes; it is doing so on a compressed timescale, fuelled by the frictionless nature of software and a venture-capital-infused culture of rapid, unchecked growth.

The data on publication volume is staggering. While medicine's "Moore's Law" posited a 12-year doubling time, the AI and ML categories on the preprint server ArXiv are expanding at a blistering pace. Annual submissions have skyrocketed from around 10,000 in 2018 to well over 50,000 in recent years.

> Some analyses suggest the number of papers in the field is doubling roughly every 23 months, let that sink in for a moment!

This is an order of magnitude faster than the growth that swamped biomedical research. The ML community is generating a body of literature so vast and so fast-moving that even dedicated practitioners cannot keep up, creating the perfect conditions for the same pathologies of noise and redundancy that now plague medicine.

A Familiar Sickness: The Reproducibility Crisis in Code

Just as in medicine, this explosion in quantity has been accompanied by a crisis in quality. ML has its own, deeply entrenched reproducibility crisis, but its causes are rooted in the specifics of software and data science. The problem is not just flawed statistical methodology but a host of technical failures: researchers failing to share their source code or the specific datasets used for training; the use of proprietary, inaccessible data; non-deterministic training processes that produce different results even on the same machine; and the insidious practice of "seed hacking," where researchers run an experiment with dozens of different random seeds and only report the one that produces the best result.

One of the most pervasive and damaging issues is data leakage. A comprehensive investigation by researchers at Princeton University compiled evidence of systemic reproducibility failures across dozens of scientific fields that use ML. They found that a common error, data leakage—where information from the test set inadvertently "leaks" into the training process—led to wildly overoptimistic and non-reproducible claims of model performance.

This is a clear echo of medicine's failures, a digital-age version of failing to properly blind a clinical trial, and it demonstrates that the cultural disregard for rigor is just as prevalent in computer science as it is in biomedicine.

The "Static Benchmark" Fallacy: A Deeper Ignorance

The methodological parallels run deeper still. If medicine is guilty of the ‘Perfect Measurement’ fallacy, then ML is guilty of its direct equivalent: the ‘Static Benchmark’ fallacy. The field's dominant paradigm for evaluating progress is to measure the performance of models on static, unchanging benchmarks. These are curated datasets like ImageNet for vision or benchmarks like the Multi-task Language Understanding (MMLU) for language models. This is as fundamentally misguided as using fMRI to understand software.

LLMs, like the human brain, are highly complex, nonlinear dynamical systems. Their behaviour is dynamical, context-dependent, sensitive to initial conditions, and emergent: the outputs evolve over time. To evaluate such a system with a multiple-choice test is a profound category error. The field relies on static tests on, for example 'intelligence' and not dynamic tests because nobody can define dynamic interaction tests or even ‘intelligence’.

It seems to me there is a failure to develop evaluation methods that can probe the dynamic, interactive, and often unpredictable nature of these models. This in turn means that the field is optimizing for a narrow, incomplete, and ultimately misleading definition of success. This means we are building systems that are good at passing tests, without a deep understanding of whether they are genuinely capable or robust.

The Missing Foundations: AGI Hype and Theoretical Voids

For me, this reliance on empirical benchmarks is a direct consequence of the field's lack of a deep theoretical foundation. My experience is that most ML practitioners ‘have never heard of Bertrand Russell, or nonlinear dynamical system theory’ is a pointed one. It highlights a culture that prioritizes engineering ‘hacks’ and scaling laws over a first-principles understanding of the systems being built: and importantly how language works and what knowledge actually means.

This theoretical void is most apparent in the confused and contradictory discourse surrounding ‘emergence’ in LLMs. As these models have scaled, they have demonstrated surprising new capabilities that were not explicitly programmed. The field has no consensus model to explain this phenomenon. One camp dismisses them as ‘stochastic parrots’, mere pattern-matchers without true understanding. Another attempts to reverse-engineer ‘token pathways’ in high-dimensional space. A third, more commercially-minded group, is quick to label any new capability as a step toward Artificial General Intelligence (AGI).

> This is a direct parallel to the decades-long over-hyping of neuroimaging in medicine, where a powerful new tool was mistaken for a solution, leading to a flood of exciting but

> ultimately shallow and often irreproducible claims about the nature of consciousness. Without a rigorous theoretical framework, a field is susceptible to hype, and ML is currently in the grip of a hype cycle of historic proportions.

The parallels between the two fields are not coincidental; they are structural. Both are grappling with systems of immense complexity, and both have defaulted to a culture that rewards the rapid production of superficially plausible results over the slow, difficult work of genuine understanding. Consider the following table:

Feature	Healthcare Research	Machine Learning Research
Publication Dynamics	"Moore's Law": ~1.3M papers/year [3] with diminishing returns on life expectancy.[5]	Exponential Growth: >50,000 AI papers/year on ArXiv [15], doubling every ~23 months.[16]
Validation Crisis	Reproducibility Crisis: >70% of researchers fail to reproduce others' work.[10]	Pervasive Irreproducibility: Driven by data leakage [20], random seeds, unshared code.[17]
Core Flaw	The "Perfect Measurement" Fallacy: Ignoring uncertainty in noisy physiological data (EEG, fMRI).[1]	The "Static Benchmark" Fallacy: Evaluating dynamic, nonlinear systems (LLMs) with simplistic, static tests.[1]
Hype Cycle Example	Using fMRI to "decode" consciousness without acknowledging severe limitations.[1]	Claiming AGI emergence from scaling laws without a foundational theory of emergence.[1]
The LLM Impact	Amplifies literature noise, making evidence-based practice harder.[1]	Automates the generation of coherent but shallow papers, accelerating the crisis.[1]

While medicine's mistakes developed over decades, constrained by the physical realities of labs and clinical trials, ML is industrializing them. The ‘move fast and break things’ ethos of Silicon

Valley is being combined with the near-zero marginal cost of replicating software. This means that flawed ideas, buggy code, and misleading benchmarks can propagate through the entire research community in months, not years. The disease that took half a century to bring medicine to a state of crisis is threatening to overwhelm ML in less than a decade. The acceleration itself is a key source of systemic risk.

> Furthermore, the reproducibility crisis and the lack of theoretical grounding in ML are two sides of the same coin. Considering the insight that LLMs are nonlinear dynamical systems is maybe a key to unlocking some of these issues.

Complex non-linear systems are characterized by their sensitivity to initial conditions, a property colloquially known as the "butterfly effect." When a field lacks a deep theoretical understanding of these dynamics, its experiments become highly susceptible to uncontrolled variables, such as the random seed used to initialize a model's weights. This extreme sensitivity is, by definition, the source of irreproducibility. The field's reproducibility crisis is therefore a direct and predictable consequence of its theoretical immaturity. It is attempting to build reliable bridges without a mastery of the underlying physics.

The Great Multiplier: LLMs as the Engine of Scientific Chaos

The convergence of these two failing systems is poised to be accelerated by the very technology that sits at the heart of the ML revolution: Large Language Models. LLMs are not merely a new tool for scientists; they are a force multiplier for the existing pathologies of the wider research ecosystem.

> It seems to me that AI and LLMs **are set to compound medical research issues of data overload and increase the noise on a scale that will make the current crisis seem quaint.**

The Idea Generator: Shiny, Coherent, and Hollow

The most immediate danger of LLMs lies in their remarkable ability to generate plausible-sounding text and ‘new ideas’. A researcher can prompt a model to devise a new hypothesis for a disease mechanism or a novel algorithm for data analysis, and in seconds, receive a polished,

well-structured, and seemingly insightful response, complete with citations and even mathematical equations.

> However, this output is generated without what can only be described as a "feel of the issues"—the deep, tacit, and hard-won understanding of a field's history, its dead ends, and its subtle complexities that only comes from years of dedicated practice.

The LLM does not have insight; it generates a "new painting," a clever recombination of patterns found in its vast training data. It is a grand master of coherence, but it is devoid of understanding. A junior researcher, already under immense pressure to publish, is presented with this seductive output. It ***looks*** right, it ***feels*** novel, and it saves months of difficult intellectual labour. The temptation to take this hollow idea and run with it is immense. And how are they also to know if these ideas are genuinely worthwhile?

The LLM as a Coherence Engine

The danger is subtle: the LLM is not a partner in thought but a **coherence engine**. It stitches together language into patterns that feel complete, even when the underlying dimensions of understanding are missing. What emerges is not a discovery but a projection of coherence into gaps: **this is a hallucination dressed in the garments of insight**. Without the grounding of genuine expertise, it becomes almost impossible to tell the difference.

The Tsunami of Noise: From 1 Million Papers to 10 Million

The consequences of this, for the signal-to-noise problem, are immense. If an LLM can draft the bulk of a research paper in hours instead of months, the current deluge of 1.3 million medical papers per year will become a tidal wave, a tsunami. The fear of an ever growing ocean of papers where people can not see the flaws in their own work is not hyperbole; it is a straightforward extrapolation.

In such an environment, the discovery of genuine breakthroughs becomes a statistical impossibility. An analogy may be painfully apt: **finding a truly significant study will be like searching for a single, specific drop of rain in a thunderstorm**. The very mechanisms of

scientific discovery from literature review, peer discussion, and the slow accumulation of consensus, will break down. They are likely to completely overwhelmed by an impenetrable fog of low-quality, AI-generated content and even AI-review as shown later.

The Great Seduction

The greatest danger here is not that LLMs produce incorrect information, but that they produce ***seductively coherent*** incorrect information. The scientific method was developed over centuries as a toolkit to combat innate human cognitive biases, such as confirmation bias and our weakness for a compelling narrative. LLMs are optimized to appeal directly to these biases. Their fluent, well-structured, and confident outputs may act as a cognitive anaesthetic, lowering the critical guards of even the most well-meaning researchers. They automate the process of fooling ourselves.

The Automation of Error: Peer Review's Final Collapse

The final line of defence against this flood of poor-quality research should be the peer review process. However, peer review is already a broken system, with overworked and often under-qualified reviewers failing to catch even major methodological flaws. The introduction of LLMs into this process as we are already seeing may well deliver the coup de grâce.

Already we are seeing that LLMs are being used to assist in or even conduct peer review, they will do exactly as they are told: they will follow the rules that they are given, but now just quicker. They will check for formatting, for the presence of the correct sections, and for grammatical coherence. What they cannot do is exercise genuine critical judgment. They cannot assess whether a study's core premise is nonsensical, whether its interpretation of the data is biased, or whether it ignores a crucial piece of contradictory prior work. Automating a broken process does not fix it; it entrenches its flaws and executes them with ruthless, machine-like efficiency. The likely result may be the rubber-stamping of coherent-but-flawed papers at an unprecedented scale.

The "Nonlinear Divergent System"

The most sophisticated and chilling prediction is that this confluence of factors will create a nonlinear divergent system that has no structure. This is a concept borrowed from the language of chaos theory, and it describes the ultimate endpoint of the crisis. The scientific literature will become a closed, self-referential feedback loop.

The process may look like this:

1. Researchers use LLMs to generate hypotheses based on the existing body of literature.
2. They conduct studies based on these plausible but shallow hypotheses and use LLMs to help write the papers.
3. These papers, having passed through a weakened, AI-assisted peer review process, are published and added to the scientific corpus.
4. The next generation of LLMs is then trained on this new, expanded corpus, which now contains a higher proportion of AI-generated, derivative, and potentially flawed content.

With each turn of this cycle, the literature becomes more and more a reflection of itself, and less and less a reflection of empirical reality. The system will begin to diverge from the real world, creating its own **internally consistent** but factually **unmoored universe of knowledge**. This represents a potential phase transition for science, a point where the cost of finding a true, externally-validated signal within the noise becomes effectively infinite. It is the intellectual equivalent of the heat death of the universe of research—a state of maximum entropy where no useful work can be done. At this point the system will be in equilibrium with its own noise and leads to a dead end. But if we understand the root of the coherence trap, we can begin to design an escape and a new path.

Finding a New Way Forward

Beyond the Illusion of Coherence

To find a way out of this crisis, one must first arrive at a deeper diagnosis. The symptoms—the paper deluge, the reproducibility crisis, the hype cycles—are clear. But what is the underlying disease? The central philosophical puzzle is this: if we looked at any individual paper it would likely seem coherent. This tells us something deeper. **This coherence is the problem**.

Each paper, in isolation, appears rational. It follows the prescribed form of scientific discourse: an introduction, a clear hypothesis, a description of methods, a presentation of results, and a

concluding discussion. It ticks all the boxes. Yet, the collective output of this process is a system that produces very little of lasting value. This surface-level rationality of the parts masks a profound irrationality of the whole. The system is a machine optimized to produce outputs that have the *appearance* of science but which often lack its essential *substance*: verifiability, novelty, and real-world impact.

Concluding Thought

The current trajectory of scientific research is leading us to a thunderstorm of noise so vast and intense that every individual drop of rain, no matter how pure or vital, is lost in the deluge.

Continuing down this path is not just inefficient; it may well be an existential threat to the scientific enterprise. The public's trust, already frayed, will not survive a system that produces millions of studies but fails to solve the problems that matter.

However, the thunderstorm of noise is not inevitable. The same tools that accelerate the crisis can be harnessed to end it. By reforming our incentives, embracing uncertainty, and thinking in terms of new systems, we can change the atmospheric conditions and calm the storm. This means we must build into the system incentives for quality, and embedding uncertainty quantification into our methods. It is also essential we develop a theoretical foundation for AI that is robust enough to prevent it from simply echoing our own noise.

> So now the task is to build a system where only the most vital drops of knowledge are identified. The stakes are nothing less than the future of truth itself. Maybe this is the biggest challenge of all in a new era of AI.

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• *Non-linear Dynamics in LLMs* • *Geofinitism* • *Finite Mechanics* •